

Question 1

Topic: Population, Fitness and Selection

Use a **Genetic Algorithm** to maximize the function:

$$f(x) = x^2$$

Where **x** ranges from 0–15.

Given Initial Population

Chromosome	Decimal Value (x)
0011	3
0101	5
1001	9
1110	14

Tasks

1. Calculate the **fitness value** $f(x) = x^2$ for each chromosome.
2. Rank the chromosomes according to **fitness value**.
3. Apply **selection** and choose the **two best chromosomes** as parents for the next generation.

Question 2

Topic: Crossover / Generating Generations

A delivery company wants to find the **shortest delivery route** between four locations.

Locations

A, B, C, D

Distance Table

From	To	Distance
A	B	10
A	C	15
A	D	20
B	C	35
B	D	25
C	D	30

Given Initial Population (Routes)

1. A → B → C → D
2. A → C → B → D
3. B → A → D → C
4. C → B → A → D

Tasks

1. Calculate the **total distance (fitness)** for each route.
 2. Select the **two best routes** as parents.
 3. Apply **single-point crossover** to generate **two new offspring routes**.
 4. Show the **new generation of routes**.
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Question 3

Topic: Mutation

An online warehouse wants to **optimize product placement** so that frequently purchased products are placed closer to the packing area.

Products and their demand levels are given as:

A (High), B (Medium), C (Low), D (High).

The given **initial population (arrangements of products)** is:

- A B C D
- B C A D
- D A B C
- C B D A

Tasks

1. Represent each product arrangement as a **chromosome**.
 2. Apply **mutation by randomly swapping two product positions** in each chromosome.
 3. Write the **new mutated population**.
 4. Briefly explain how mutation helps in finding **better product placement**.
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Question 4

Topic: Problem Solving using Genetic Algorithm

A factory wants to determine the best values of **x, y, and z** that satisfy the equation:

$$6x^3 + 9y^2 + 90z = 256x^3 + 9y^2 + 90z = 25$$

The **initial population of possible solutions** is given as:

- (1, 1, 0)
- (2, 1, 0)
- (1, 2, 0)
- (2, 2, 0)

The **fitness function** is:

$$\text{fitness} = \frac{1}{|6x^3 + 9y^2 + 90z - 25|}$$

Tasks

1. Evaluate the expression **$6x^3 + 9y^2 + 90z$** for each solution.
2. Calculate the **fitness value** for each solution.
3. Select the **two best solutions based on highest fitness**.
4. Apply **mutation by slightly changing the values of x, y, or z (± 0.1)** to generate a **new population**.
5. Repeat the process for several generations to approach the **target value 25**.